

Row IV: evolution, DGLAP and finite

The plus prescription in ξ , and the exact three-way split of the result

$$\xi = \frac{k^+}{k^+ + p^+}, \quad \bar{\xi} = 1 - \xi, \quad \xi_0 = \frac{k^+}{\sqrt{V}}, \quad r = y' - y, \quad \kappa = k_\perp \cdot r, \quad X = \frac{x - y}{(x - y)^2}, \quad X' = \frac{x' - y'}{(x' - y')^2}, \quad \hat{r} = \frac{r}{|r|}, \quad L_r = \log(4\pi^2 \mu^2 r^2)$$

Yes, and the split is exact. Writing $\frac{1}{1-\xi} = \left[\frac{1}{1-\xi}\right]_+ + \delta(1-\xi) \log \frac{k^+}{\Lambda}$ separates your (0.24) into three pieces that sum back to it identically (checked to 10^{-14} at three kinematic points):

$$\frac{d^3 N^{\text{NLO}}}{d^3 k} = \underbrace{\log \frac{k^+}{\Lambda} \otimes \text{BK kernel}}_{\text{evolution}} + \underbrace{\frac{1}{\bar{\xi}} \otimes \hat{P}_{gg}(\xi)}_{\text{DGLAP}} + \underbrace{\text{NLO coefficient function}}_{\text{finite}}$$

0. Status of the note

Pages 1–3 are unchanged except that (0.20) now reads $\frac{\pi \delta_{mm'}}{d_\perp} \left[\frac{1}{\bar{\xi}} + 1 - \log(4\pi^2 \mu^2 r^2) \right] - \frac{\pi r^m r^{m'}}{r^2}$ with $\frac{1}{\bar{\xi}} = \frac{1}{\xi} - \gamma_E + \log 4\pi$ — **correct**, and the plain μ is now right. The new page 4, eq. (0.24), is **correct**: signs, the $\frac{1}{\xi^2}$ Jacobian on both blocks, the phase $e^{-ik \cdot r / \xi}$ and the splitting function all check out.

Still to fix, all of it bookkeeping outside the ξ algebra:

where	what
(0.2)	the overall $\frac{(k^+ + p^+)}{k^+}$ is spurious (prefactor is $\frac{1}{k^+}$), and the $\frac{S}{k^+}$ belongs in the exponent — (0.3) already has it right
(0.18), (0.19)	$\mu^{2e} \rightarrow \mu^{-2e}$, otherwise (0.20) does not follow (and the log argument is dimensionful)
$\mathfrak{C}_{1jki}$	$\Theta(k_{\text{arg}}^+ - p_{\text{arg}}^+ - \Lambda) = \Theta(k^+ - \Lambda)$ here, so it does not supply the $\xi \rightarrow 1$ cutoff the plus prescription needs
(0.3)–(0.24)	$\int_{z,w,w'} \rightarrow \int_z$ then delete; $\int_{x,y,y'} \rightarrow \int_{x,x',y,y'}$; $U^{ce}(x) \rightarrow U^{ec}(x)$
page 3	the “Expanding the gamma functions” paragraph is printed twice; (0.10)/(0.11) carry a stray $x(1-x)$ in the denominator of the last term

1. The row, ascending in ξ

Flipping your limits (and the signs with them) so nothing has to be tracked:

$$\begin{aligned} \frac{d^3 N^{\text{NLO}}}{d^3 k} &= \mathcal{P} \int_{\xi_0}^{\xi_1} \frac{d\xi}{2\pi} \frac{1}{\xi^2} \int_{x,x',y,y'} e^{-i\kappa/\xi} \left\{ \mathcal{C}(\xi) \mathcal{M} \mathcal{X} - d_\perp \xi \bar{\xi} \mathcal{R} - \left[\frac{\xi}{\xi} + \frac{\bar{\xi}}{\xi} \right] \mathcal{X} \right\} \mathcal{U} \\ \mathcal{P} &= \frac{1}{(2\pi)^3} \frac{g^4}{4\pi^3} \frac{N_c}{k^+}, \quad \mathcal{C}(\xi) = \xi \bar{\xi} + \frac{\bar{\xi}}{\xi} + \frac{\xi}{\bar{\xi}} = \frac{P_{gg}(\xi)}{2N_c}, \quad \mathcal{M} = \frac{1}{\bar{\xi}} + 1 - L_r \\ \mathcal{X} &= (X \cdot X'), \quad \mathcal{R} = (X \cdot \hat{r})(X' \cdot \hat{r}), \quad \xi_1 = \frac{k^+}{k^+ + \Lambda}, \quad \mathcal{U} = [U^{e'c}(y') - U^{e'c}(x')] [U^{ec}(y) - U^{ec}(x)] \rho^{e'}(x') \rho^e(x) \end{aligned}$$

2. Isolating the $\xi \rightarrow 1$ pole

Only $\frac{\xi}{\xi}$ is singular, and it appears twice — once inside $\mathcal{C}$ multiplied by $\mathcal{M}$, once in the last block multiplied by -1 . They combine:

$$\frac{1}{\xi^2} e^{-i\kappa/\xi} \left[\mathcal{M} - 1 \right] \frac{\xi}{1-\xi} \mathcal{X} = \frac{h(\xi)}{1-\xi} \left[\mathcal{M} - 1 \right] \mathcal{X}, \quad h(\xi) \equiv \frac{e^{-i\kappa/\xi}}{\xi}$$

and $h(1) = e^{-i\kappa}$ is finite, so the prescription is legitimate:

$$\int_{\xi_0}^{\xi_1} d\xi \frac{h(\xi)}{1-\xi} = h(1) \log \frac{1-\xi_0}{1-\xi_1} + \int_{\xi_0}^{\xi_1} d\xi \frac{h(\xi) - h(1)}{1-\xi} \xrightarrow{\Lambda \ll k^+ \ll V} e^{-i\kappa} \log \frac{k^+}{\Lambda} + \int_{\xi_0}^1 d\xi \frac{h(\xi) - h(1)}{1-\xi}$$

Everything else — $\xi \bar{\xi}$, $\frac{\bar{\xi}}{\xi}$, $\mathcal{M}$, $\mathcal{X}$, $\mathcal{R}$ and the Wilson lines — is regular at $\xi = 1$.

Regularised splitting function.

$$\hat{C}(\xi) \equiv \xi \bar{\xi} + \frac{\bar{\xi}}{\xi} + \frac{\xi}{[1-\xi]_+} = \frac{\hat{P}_{gg}(\xi)}{2N_c}, \quad \int_{\xi_0}^1 d\xi \frac{e^{-i\kappa/\xi}}{\xi^2} \frac{\xi}{[1-\xi]_+} \equiv \int_{\xi_0}^1 d\xi \frac{h(\xi) - h(1)}{1-\xi}$$

3. The three sectors

Substituting and regrouping by what absorbs each piece (write $\mathcal{P}_0 \equiv \frac{1}{(2\pi)^3} \frac{g^4}{8\pi^4} \frac{N_c}{k^+}$, the $\frac{1}{2\pi}$ of the measure already included):

(A) Evolution sector

$$\left. \frac{d^3 N^{\text{NLO}}}{d^3 k} \right|_{\text{evol}} = \mathcal{P}_0 \log \frac{k^+}{\Lambda} \int_{x,x',y,y'} e^{-ik_{\perp} \cdot (y'-y)} \left[\frac{1}{\bar{\epsilon}} - \log(4\pi^2 \mu^2 (y'-y)^2) \right] \frac{(x-y) \cdot (x'-y')}{(x-y)^2 (x'-y')^2} \mathcal{U}$$

The bracket is exactly $\frac{1}{\pi} \delta_{mm'} \mathcal{G}^{mm'}$, so this is the z integral put back:

$$= \frac{1}{(2\pi)^3} \frac{g^4}{8\pi^5} \frac{N_c}{k^+} \log \frac{k^+}{\Lambda} \int_{x,x',y,y'} e^{-ik_{\perp} \cdot (y'-y)} \frac{(x-y) \cdot (x'-y')}{(x-y)^2 (x'-y')^2} \int d^2 z \frac{(y-z) \cdot (y'-z)}{(y-z)^2 (y'-z)^2} \mathcal{U}$$

Note the hard scale is $k_{\perp}$, not $k_{\perp}/\xi$: at $\xi = 1$ nothing is emitted into the measured direction. This is the real-emission piece of the JIMWLK/BK kernel multiplied by the rapidity logarithm, i.e. $\partial_{\log(1/\Lambda)}$ of the dipole. Its $\frac{1}{\bar{\epsilon}}$ is the large- $|z|$ end of that kernel and cancels once the companion rows supply the $\frac{1}{(y-z)^2}$ and $\frac{1}{(y'-z)^2}$ pieces that complete $\frac{(y-y')^2}{(y-z)^2 (y'-z)^2}$.

(B) DGLAP sector

$$\left. \frac{d^3 N^{\text{NLO}}}{d^3 k} \right|_{\text{DGLAP}} = \frac{1}{(2\pi)^3} \frac{g^4}{8\pi^4} \frac{1}{k^+} \frac{1}{\bar{\epsilon}} \int_{\xi_0}^1 \frac{d\xi}{\xi^2} \frac{\hat{P}_{gg}(\xi)}{2} \int_{x,x',y,y'} e^{-i\frac{k_{\perp}}{\xi} \cdot (y'-y)} \frac{(x-y) \cdot (x'-y')}{(x-y)^2 (x'-y')^2} \mathcal{U}$$

$$\hat{P}_{gg}(\xi) = 2N_c \left[\xi(1-\xi) + \frac{1-\xi}{\xi} + \frac{\xi}{[1-\xi]_+} \right]$$

Written with the LO structure $\mathcal{N}_{\text{LO}}(q_{\perp}) = \int_{x,x',y,y'} e^{-iq_{\perp} \cdot (y'-y)} \frac{(x-y) \cdot (x'-y')}{(x-y)^2 (x'-y')^2} \mathcal{U}$ this is

$$= \frac{\alpha_s}{2\pi} \frac{1}{\bar{\epsilon}} \int \frac{d\xi}{\xi^2} \frac{\hat{P}_{gg}(\xi)}{2} \frac{d^3 N^{\text{LO}}}{d^3 k} \left(\frac{k_{\perp}}{\xi} \right)$$

— textbook collinear factorisation for fragmentation. Remove it with

$$D_{g/g}(\xi, \mu_F) = \delta(1-\xi) + \frac{\alpha_s}{2\pi} \left[-\frac{1}{\bar{\epsilon}} + \log \frac{\mu_F^2}{\mu^2} \right] \hat{P}_{gg}(\xi) + O(\alpha_s^2)$$

which leaves $\mu \rightarrow \mu_F$ in the finite sector.

(C) Finite sector

$$\left. \frac{d^3 N^{\text{NLO}}}{d^3 k} \right|_{\text{fin}} = \mathcal{P}_0 \int_{\xi_0}^1 \frac{d\xi}{\xi^2} \int_{x,x',y,y'} e^{-i\frac{k_{\perp}}{\xi} \cdot (y'-y)} \left\{ -\log(4\pi^2 \mu_F^2 r^2) \hat{C}(\xi) (X \cdot X') \right.$$

$$\left. + \xi(1-\xi) \left[(X \cdot X') - d_{\perp}(X \cdot \hat{r})(X' \cdot \hat{r}) \right] \right\} \mathcal{U}$$

Two structures only. The first is the same $\hat{P}_{gg}$ now riding on the position-space logarithm $\log(4\pi^2 \mu_F^2 r^2)$ — the NLO hard factor, which does *not* factor out of $\mathcal{N}_{\text{LO}}$ because r is integrated. The second is the pure transverse tensor

$$X^m X^{m'} \left[\delta^{mm'} - d_\perp \hat{r}^m \hat{r}^{m'} \right] \xi (1 - \xi)$$

which has no pole, no $\frac{1}{1-\xi}$ and no log: it is the azimuthal partner of the $\xi \bar{\xi}$ term of P_{gg} , and it vanishes on averaging over $\hat{r}$ at $d_\perp = 2$.

4. The decomposition is exact

Sum of the three sectors against the exact (0.24), at three kinematic points (with the upper limit kept at ξ_1 throughout, so no $O(\Lambda/k^+)$ is dropped):

(k^+, κ)	exact (0.24)	evol + DGLAP + fin	diff
(1.0, +0.9)	$826.912590 + 338.307363 i$	$826.912590 + 338.307363 i$	2.7×10^{-14}
(2.5, -1.4)	$-150.161862 + 96.165817 i$	$-150.161862 + 96.165817 i$	4.5×10^{-15}
(0.7, +2.2)	$-36.595454 + 176.960721 i$	$-36.595454 + 176.960721 i$	2.8×10^{-14}

Replacing $\log \frac{1-\xi_0}{1-\xi_1}$ by $\log \frac{k^+}{\Lambda}$ and the upper limit ξ_1 by 1 costs $O(\frac{\Lambda}{k^+})$ — e.g. 2×10^{-3} out of 8×10^2 at $\Lambda/k^+ = 10^{-4}$, scaling linearly in Λ as it must.

5. Final answer

$$\boxed{\frac{d^3 N^{\text{NLO}}}{d^3 k} \Big|_{\text{Row IV}} = \frac{d^3 N}{d^3 k} \Big|_{\text{evol}} + \frac{d^3 N}{d^3 k} \Big|_{\text{DGLAP}} + \frac{d^3 N}{d^3 k} \Big|_{\text{fin}}}$$

$$\frac{d^3 N}{d^3 k} \Big|_{\text{evol}} = \frac{1}{(2\pi)^3} \frac{g^4}{8\pi^5} \frac{N_c}{k^+} \log \frac{k^+}{\Lambda} \int_{x,x',y,y'} e^{-ik_\perp \cdot r} \frac{(x-y) \cdot (x'-y')}{(x-y)^2 (x'-y')^2} \int d^2 z \frac{(y-z) \cdot (y'-z)}{(y-z)^2 (y'-z)^2} \mathcal{U}$$

$$\frac{d^3 N}{d^3 k} \Big|_{\text{DGLAP}} = \frac{1}{(2\pi)^3} \frac{g^4}{8\pi^4} \frac{1}{k^+} \frac{1}{\bar{\epsilon}} \int_{\xi_0}^1 \frac{d\xi}{\xi^2} \frac{\widehat{P}_{gg}(\xi)}{2} \int_{x,x',y,y'} e^{-i\frac{k_\perp}{\xi} \cdot r} \frac{(x-y) \cdot (x'-y')}{(x-y)^2 (x'-y')^2} \mathcal{U}$$

$$\frac{d^3 N}{d^3 k} \Big|_{\text{fin}} = \frac{1}{(2\pi)^3} \frac{g^4}{8\pi^4} \frac{N_c}{k^+} \int_{\xi_0}^1 \frac{d\xi}{\xi^2} \int_{x,x',y,y'} e^{-i\frac{k_\perp}{\xi} \cdot r} \left\{ -\log(4\pi^2 \mu_F^2 r^2) \hat{\mathcal{C}}(\xi) \mathcal{X} + \xi \bar{\xi} [\mathcal{X} - d_\perp \mathcal{R}] \right\} \mathcal{U}$$

6. Three things to keep in view

- **The evolution sector still carries $\frac{1}{\bar{\epsilon}}$.** That is not a defect of this row: it is the unscreened large- $|z|$ end of the BK kernel, and it only cancels when the Group II rows in which $U(z)$ survives the colour reduction are added. Do not absorb it into $D_{g/g}$ — a fragmentation function cannot depend on Λ , and this term does.
- **The small- ξ end.** $\widehat{P}_{gg} \rightarrow \frac{2N_c}{\xi}$ against $\frac{d\xi}{\xi^2}$ gives a ξ^{-3} integrand at $\xi \rightarrow \xi_0 = \frac{k^+}{\sqrt{V}}$, controlled only by the oscillation of $e^{-ik_\perp \cdot r/\xi}$. Do the y, y' integrals before sending $V \rightarrow \infty$.
- **The Λ in $\log \frac{k^+}{\Lambda}$** is not the one supplied by the Θ in your $\mathfrak{C}_{1jki}$, which reduces to $\Theta(k^+ - \Lambda)$ here. Settle where the $\xi \rightarrow 1$ cutoff comes from before quoting this coefficient against the other rows.

Verified in `src/rowG2_sectors.py` (the three-sector sum against the exact ξ integrand), `src/rowG2_dglap.py` (the P_{gg} identity and the $\frac{\alpha_s}{2\pi}$ normalisation), `src/rowG2_xicheck.py` (the change of variable) and `src/rowG2_vertex.py` (eq. (0.1) from the vertex definitions).